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Teaching an LLM to Read Emotions in a Language None of Us Speaks

Large Language Models

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Contents

1 Introduction and Motivation **4**

2 Background **4**
2.1 Multi-label Emotion Classification 4
2.2 XED, Label Projection and Cross-Lingual Transfer 5
2.3 Parameter-Efficient Fine-Tuning 5
2.4 Prompting 5

3 Data and Task Setup **5**
3.1 XED for English and Croatian 5
3.2 Split and Leakage Controls 6
3.3 Data Formatting 6

4 Methodology **7**
4.1 Models 7
4.2 Parameter-Efficient Fine-Tuning 7
4.3 Prompting 8
4.4 Output Parsing 8

5 Evaluation **9**
5.1 Metrics 9
5.2 Protocol 9

6 Results **9**
6.1 PEFT and Prompting 9
6.2 Seeds and Learning Curve 10
6.3 Per-Emotion Results 11

7 Analysis and Discussion **12**
7.1 Cross-Lingual Transfer 12
7.2 Label Counts and Output Format 12
7.3 Per-Emotion Behaviour 13
7.4 Prompt Sensitivity 13
7.5 PEFT versus Prompting: Cost 14
7.6 Error Analysis 14
7.7 Limitations 15

8 Conclusion and Future Work **16**

A Code Repository **18**

B Language Resource Bands **18**

C Split Statistics **19**

D Prompts **19**

E Fine-Tuning Configuration **21**
E.1 Configuration Search 21
E.2 Training Histories 23

F Prompting on the Development Set **25**

G Further Test Results **27**

1 Introduction and Motivation

Emotion detection asks which emotions a piece of text expresses. Dialogue makes this harder, as a movie subtitle is short, carries almost no context, and often conveys emotion implicitly through irony, a question, or a reaction to something the reader cannot see. A line can also express several emotions at once; "You did what?" may be read as both surprise and anger. The task is therefore multi-label, in that a classifier must decide, for each of a fixed set of emotions, whether it is present, so partially correct answers are common and rare emotions are hard both to learn and to evaluate.

Large language models (LLMs) offer two ways to approach this task. They can be prompted to label emotions directly, from a description of the task and optionally a few solved examples, or, when labelled data is available, adapted to it by training a small number of additional weights. Whether either works for a fine-grained, multi-label task beyond English is less clear: LLMs see far less text in most other languages during pretraining, and labelled data outside English is scarce. In the XED dataset (Öhman et al., 2020), only English and Finnish were annotated by people, and the labels for other languages were projected onto aligned subtitles, so any evaluation in these languages inherits their noise.

In this project, we study how well open-weight LLMs classify emotions in English and in Croatian, our assigned language, comparing prompt engineering, with zero or five examples in the prompt, and parameter-efficient fine-tuning (PEFT). Training on English alone and evaluating on Croatian also shows how much of what is learned transfers across languages. Croatian falls in band 3 of the project's five resource bands ("mid-low", roughly 4,400–6,000 annotated lines; Table 5). No member of our group reads Croatian, so Croatian outputs are judged only against XED's projected labels. We:

- build a fixed, leakage-controlled split of matched English–Croatian pairs;
- fine-tune our primary model, EuroLLM-9B, on English, Croatian and both, with one configuration selected on the development set, and retrain on smaller Croatian samples to see how much target-language data fine-tuning needs;
- prompt EuroLLM-9B and a second model, Qwen3.5-9B, zero-shot and five-shot, and measure sensitivity to prompt wording; and
- evaluate every condition with multi-label metrics and bootstrap confidence intervals, and analyse concrete errors, separating model errors from those that may stem from subtitle alignment or label projection.

Section 2 introduces multi-label emotion classification, XED and its projected labels, and the two adaptation approaches. Section 3 describes the data and split, Section 4 the models, fine-tuning and prompts, and Section 5 the evaluation protocol. Sections 6 and 7 present and discuss the results, and Section 8 concludes. The code and configuration files are available at https://github.com/squidistaken/large-language-models.

2 Background

2.1 Multi-label Emotion Classification

XED (Öhman et al., 2020) uses Plutchik's eight basic emotions: anger, anticipation, disgust, fear, joy, sadness, surprise, and trust. Since a subtitle line can carry several of these emotions, a model has to make eight separate yes or no decisions rather than select one emotion class. We will therefore

score each prediction as a set, mainly with micro-averaged F1 and per-line Jaccard similarity, because predictions are rarely all correct or all wrong, which plain accuracy does not capture. Emotions are also unevenly spread, so rare ones leave little to learn from.

2.2 XED, Label Projection and Cross-Lingual Transfer

XED is a dataset of movie subtitles from OpenSubtitles, in which each line comes without the dialogue surrounding it. Only the English and Finnish lines were annotated by people. Labels for thirty additional languages, including Croatian, were obtained by annotation projection, and we will treat them as projected reference labels rather than human judgements. In annotation projection, the subtitles of the same film in two languages are aligned, mainly by their timing, and the emotions of a line are copied to its aligned line. A projected label may therefore be inaccurate, because aligned lines are not necessarily translations of each other, or because the translation changed the emotion of the original line.
Cross-lingual transfer means training in one language and testing in another. We will test it by evaluating an English-only fine-tuned model on Croatian, and a bilingual condition will show whether adding Croatian training data helps beyond that.

2.3 Parameter-Efficient Fine-Tuning

Updating every weight of a model with billions of parameters is costly, since training also keeps a gradient and optimiser state for each of them. Parameter-efficient fine-tuning (PEFT) avoids this by freezing the pretrained model and training only a small number of added parameters. We will compare two such methods. Low-Rank Adaptation (LoRA; Hu et al., 2022) gives each frozen weight matrix an update formed by multiplying two small trained matrices, called an adapter, which together take up a fraction of a percent of the model's parameters. QLoRA (Dettmers et al., 2023) additionally stores the frozen model in a 4-bit format, cutting the memory taken by its weights roughly fourfold at the cost of some rounding error.

2.4 Prompting

Instruction-tuned language models are trained to respond to a request described in their input, the prompt. In zero-shot prompting, the prompt holds only the instruction; in few-shot prompting, it also contains a handful of solved examples (demonstrations), from which a model can infer the task without changes to its weights (Brown et al., 2020). Prompting needs no training, but its results can be fragile, so we will analyse prompt sensitivity on the development set. Few-shot performance changes with the wording of the prompt and with which demonstrations are shown, partly because models favour labels that are frequent among the demonstrations (Zhao et al., 2021).

3 Data and Task Setup

3.1 XED for English and Croatian

We used the English and Croatian portions of XED, keeping Plutchik's eight labels, in a fixed order, from data preparation through to evaluation. Neutral lines are not part of the task, so every line carries at least one emotion. To build matched pairs, we linked every Croatian line to an English line through

XED's subtitle alignments. Of the 5,422 pairs left after cleaning, 2,236 are aligned directly (one hop). For the other 3,186, XED links the Croatian line only to a Danish subtitle, and we reached the English line through Danish (two hops). Where a Croatian line had several English counterparts, we preferred one with a human annotation, then the one with the fewest hops. The English side carries XED's human labels for 5,247 pairs (96.8%); the other 175 English lines take the projected labels of their Croatian counterpart. Since the two languages' labels come from different files, they can differ within a pair. For every line, we recorded the number of hops and the source of its labels, which we later use to separate model errors from alignment and projection errors.

Anger and anticipation are the most frequent emotions, each on about a quarter of the lines, and disgust is the rarest, on about one line in eight; a Croatian line carries 1.45 emotions on average (Figure 4).

3.2 Split and Leakage Controls

XED's Croatian file has 5,941 labelled lines once three malformed rows are removed. Of these lines, one could not be aligned to any English subtitle. We detected duplicates after normalising Unicode, case, punctuation and spacing, which differ between XED files, and kept only the first of 518 repeated Croatian lines. This leaves 5,422 English–Croatian pairs. English lines can still repeat, because different Croatian lines are sometimes aligned with the same English text (612 English lines occur in more than one pair), so we grouped the pairs that share an English line and assigned each group to a single partition. No line, in either language, therefore appears in more than one partition.

We made one fixed 70/20/10 training, development and test split with seed 42. Pairs were assigned by their XED alignment identifier before the English and Croatian files were written, which keeps both versions of a subtitle in the same partition. To keep each emotion's share similar across partitions, we used iterative multi-label stratification (Sechidis et al., 2011), as implemented in `iterstrat`, on the Croatian labels merged over each group, since every pair has a Croatian label set, whereas 175 English lines borrow theirs. The training set was split off first, and the remaining 30% was divided 2:1 into development and test sets. This gave 3,803 training, 1,068 development and 551 test lines per language, with each emotion's share differing by at most 2.3 percentage points between the Croatian partitions and 2.7 between the English ones (Table 6). All design decisions were made on the development set, the test set was used only for the final evaluation, and the five-shot demonstrations were drawn from the training set only.

3.3 Data Formatting

Every pair was written out in a conversational format of three messages. The system message lists the eight permitted labels in the fixed order, each with a one-line definition, states that every line expresses at least one of them and may express several, and asks for a JSON list of the matching label names and nothing else. The user message holds the subtitle, as `Subtitle: {text}`, and the assistant message holds the reference labels as a JSON list, for example `["anger", "fear"]`. Croatian examples use the same English instruction and label names, giving both languages the same output format, parser and metrics. Fine-tuning trains the model to produce the assistant message, while prompting lets it generate the message itself. With the same files serving both approaches, zero-shot prompting sees the same input as the fine-tuned models.

4 Methodology

We used one primary model for all prompting and fine-tuning conditions, keeping the choice of model out of the comparisons, and a second model for prompting only.

4.1 Models

We chose both models for their size, multilingual coverage, licence and accessibility: both are open-weight, released under the Apache 2.0 licence on the Hugging Face Hub, and fit on a single NVIDIA A100 (40 GB) of the university's high-performance computing cluster. All experiments used PyTorch 2.14.0, Transformers 5.17.0 and PEFT 0.21.2. The primary model is EuroLLM-9B-Instruct (Martins et al., 2025), hereafter EuroLLM-9B (`utter-project/EuroLLM-9B-Instruct`, revision `f7ae2bc3`). It has 9.15B parameters and was pretrained on about four trillion tokens in 35 languages, including all official EU languages, so Croatian is one of the languages it was built for. Prompting and fine-tuning used the same 4-bit version of it, leaving the way the model is adapted as the only difference between the two approaches. The second model is Qwen3.5-9B (Qwen Team, 2026) (`Qwen/Qwen3.5-9B`, revision `c2022362`), a widely used open-weight model of the same size; its model card claims support for "201 languages and dialects" but does not list them, so we do not know whether Croatian was deliberately covered. As it was never fine-tuned, it had no 4-bit runs to match, and we ran it at its default bfloat16 (bf16) precision, which fits on the A100. The two models were thus prompted at different precisions. We disabled its reasoning mode to make it answer with the label list directly.

4.2 Parameter-Efficient Fine-Tuning

We fine-tuned the primary model in three conditions:

1. **PEFT-en** on the 3,803 English training lines, evaluated on English and Croatian;
2. **PEFT-hr** on 3,802 of the 3,803 Croatian training lines (one exceeded the 256-token limit), evaluated on Croatian;
3. **PEFT-bilingual** on both (7,605 lines), evaluated on English and Croatian.

Each example is a three-message conversation, and the cross-entropy loss is computed on the tokens of the assistant's answer only, with the system and user messages serving as context. We used LoRA, in which an adapted layer computes

$$\mathbf{h} = W_0\mathbf{x} + \frac{\alpha}{r}BA\mathbf{x}, \tag{1}$$

where $\mathbf{x} \in \mathbb{R}^k$ is the layer's input and $\mathbf{h} \in \mathbb{R}^d$ its output, $W_0 \in \mathbb{R}^{d\times k}$ the frozen pretrained weight matrix, $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$ the trained adapter matrices of rank $r \ll \min(d,k)$, and α a scaling constant. B is initialised to zero, so training starts from the unchanged model. With QLoRA, W_0 is stored in the 4-bit NormalFloat (NF4) format with double quantisation and converted back to bf16 only when each layer is computed.

All three conditions shared one configuration: QLoRA with adapters on all linear layers, rank $r = 16$, $\alpha = 32$ and dropout 0.05, giving 50.9M trainable parameters (0.55% of the model). We fixed the rank, α and dropout rather than tuning them, as the project brief does not require a full hyperparameter grid. We trained with AdamW without weight decay, a learning rate of 1×10^{-4} with linear warm-up over the first 3% of steps and linear decay after, a batch size of 8 with two gradient-accumulation

steps (an effective batch of 16), a maximum sequence length of 256 tokens, at most three epochs, and seed 42, with gradient checkpointing to save memory and deterministic training to make runs exactly repeatable. The 256-token limit fits every training example except the one Croatian line (259 tokens). Since validation loss does not measure whether the generated label set is correct, we selected checkpoints by development micro-F1: twice per epoch, the model generated answers for every development line in its training language(s), and we kept the best checkpoint and stopped after two evaluations without improvement. The kept checkpoints are from epoch 1.5 for PEFT-en, 3.0 for PEFT-hr and 0.5 for PEFT-bilingual.

This configuration won a small development-set search over LoRA (bf16) and QLoRA (4-bit) at learning rates of 1, 2 and 4×10^{-4}. Each candidate was trained in all three conditions and scored by its development micro-F1 averaged over the conditions evaluated in each language, then over the two languages, so that Croatian, evaluated in three conditions, does not outweigh English, evaluated in two. All six candidates lay within 0.01 micro-F1 of each other (Table 8). To see how much Croatian training data fine-tuning needs, we also retrained PEFT-hr and PEFT-bilingual with the Croatian training lines capped at 250, 500, 1,000 and 2,000, keeping every other setting unchanged.

4.3 Prompting

Both models were prompted zero-shot and five-shot with the same instruction and message format as the fine-tuning data. The five-shot prompt adds five demonstrations as user and assistant turns before the test line, with the same five alignment IDs throughout: English evaluation shows the English lines with their gold labels, and Croatian evaluation the aligned Croatian lines with their projected labels (Table 7). Since few-shot results depend on the demonstrations, we chose them from the training split by a fixed procedure: candidates needed gold English labels, for human-labelled English demonstrations; a direct English–Croatian alignment, making each Croatian demonstration the closest counterpart of its English line; at most 80 characters in both languages, which keeps the prompt short and still admits about 95% of the lines; and at most three labels, as 98% of the lines have. From these, we greedily picked the candidate that added the most emotions not yet covered, one per film for variety, breaking ties at random (seed 42).

All answers were decoded greedily, which makes them deterministic, with at most 32 new tokens, enough for a list of all eight labels (31 tokens for EuroLLM-9B and 30 for Qwen3.5-9B). To measure prompt sensitivity, we also scored two reworded instructions and one set of five randomly drawn demonstrations on the development set only; only the default prompt was run on test.

4.4 Output Parsing

A single parser trims the surrounding whitespace of every answer and decodes it as JSON; an answer that does not decode to a list is counted as malformed. From a list, it keeps the string items, lowercases them and collapses their whitespace, then drops duplicates and anything that is not one of the eight permitted labels, mapping no synonyms. Malformed answers and answers with no permitted label both count as an empty prediction, and the malformed rate is reported separately. The label set is then binarised into an eight-dimensional vector in the fixed label order.

5 Evaluation

5.1 Metrics

Every metric is computed from the same binarised predictions with scikit-learn 1.9.1 (Pedregosa et al., 2011). We report two primary metrics. Micro-averaged F1 pools the counts of all eight emotions before computing F1, so frequent emotions weigh more:

$$\text{Micro-F1} = \frac{2\,\text{TP}}{2\,\text{TP} + \text{FP} + \text{FN}}, \tag{2}$$

where TP, FP, and FN are the numbers of true positives, false positives, and false negatives summed over all emotions and lines. The sample-averaged Jaccard index scores each line by the overlap between its reference and predicted label sets, giving partial credit to a line whose labels are partly right:

$$\text{Jaccard} = \frac{1}{N} \sum_{i=1}^{N} \frac{|Y_i \cap \hat{Y}_i|}{|Y_i \cup \hat{Y}_i|}, \tag{3}$$

where N is the number of lines, and Y_i and $\hat{Y}_i$ are the reference and predicted label sets of line i.
As secondary metrics, we report macro-averaged F1, the unweighted mean of the per-emotion F1 scores; micro-averaged precision and recall, which show whether a system predicts too many emotions or too few; and the Hamming loss, the fraction of all label decisions that are wrong (lower is better). As diagnostics, we report each emotion's precision, recall, F1 and support, the exact-match ratio (the share of lines whose predicted set equals the reference), and the malformed and empty rates. When a denominator is zero, the metric is set to 0.0. Since every reference line carries a label, this only affects an emotion that a system never predicts.
We give 95% confidence intervals for micro-F1, Jaccard and macro-F1 using a percentile bootstrap over lines, with 1,000 resamples (seed 42). These reflect only test-set sampling; for the variation between training runs, we repeated each PEFT condition with seeds 42, 67, and 128. Every other run used seed 42.

5.2 Protocol

A single script scores every system on the test set. The comparisons are controlled as far as the design allows: every system answers the same 551 test lines per language, with the same instruction, greedy decoding, parser and metrics; prompting and PEFT use the same 4-bit EuroLLM-9B weights; and the three PEFT conditions differ only in their training data. The English and Croatian test sets hold aligned versions of the same subtitles, so a difference between the languages does not come from evaluating on different lines, although it can come from the projected labels. The comparison between the two prompted models is the exception, since they ran at different precisions.

6 Results

6.1 PEFT and Prompting

Table 1 gives the test results of every system. In English, both fine-tuned conditions outperformed prompting the same model: PEFT-en reached a micro-F1 of 0.517 and PEFT-bilingual 0.511, against 0.428 for zero-shot and 0.425 for five-shot prompting, and the two fine-tuned conditions lie within

each other's bootstrap intervals. In Croatian, PEFT-bilingual scored highest of the EuroLLM-9B systems (0.436), ahead of PEFT-hr (0.403), while PEFT-en (0.385) scored about the same as zero-shot prompting (0.380). Every system scored lower on Croatian than on English, by between 0.030 (five-shot prompting) and 0.132 (PEFT-en) in micro-F1.

Qwen3.5-9B outperformed prompted EuroLLM-9B in every setting and metric. With five demonstrations, it reached a micro-F1 of 0.513 in English and 0.444 in Croatian, close to the best fine-tuned conditions, although its Jaccard stayed below theirs. For both models, five-shot prompting raised or kept micro-F1 but lowered Jaccard. Only one answer on the whole test set was malformed, from zero-shot EuroLLM-9B in English (0.2%).

The secondary metrics show how the systems differ. Five-shot prompting raised recall for both models in both languages, mostly at the cost of precision, and matched the full label set on only 9–14% of lines, against 22–26% for fine-tuning. Zero-shot Qwen3.5-9B was the most conservative system, with the highest precision and the lowest Hamming loss in both languages. Among the EuroLLM-9B systems, fine-tuning gave the lowest Hamming loss, and on Croatian, PEFT-en and PEFT-hr favoured precision over recall, whereas zero-shot prompting favoured recall. In Croatian, five-shot EuroLLM-9B scored above PEFT-hr on macro-F1 (0.386 against 0.380) but below it on micro-F1, so macro-F1 changes the order only of close systems.

Table 1: Test results, with 95% bootstrap intervals for the primary metrics; the best micro-F1 per language in bold. P and R are micro-averaged precision and recall, HL the Hamming loss (↓) and EM the exact-match ratio. PEFT rows use EuroLLM-9B.

System	micro-F1	Jaccard	macro-F1	P	R	HL	EM
English							
EuroLLM-9B, zero-shot	0.428 [.398, .456]	0.357 [.327, .386]	0.412	0.413	0.444	0.216	0.176
EuroLLM-9B, five-shot	0.425 [.396, .451]	0.327 [.301, .353]	0.415	0.365	0.508	0.250	0.093
PEFT-en	**0.517** [.488, .545]	0.441 [.410, .470]	0.508	0.510	0.526	0.178	0.256
PEFT-bilingual	0.511 [.483, .541]	0.451 [.420, .481]	0.494	0.504	0.518	0.180	0.263
Qwen3.5-9B, zero-shot	0.484 [.454, .513]	0.417 [.386, .450]	0.459	0.540	0.438	0.170	0.263
Qwen3.5-9B, five-shot	0.513 [.485, .539]	0.408 [.381, .435]	0.505	0.447	0.602	0.208	0.143
Croatian							
EuroLLM-9B, zero-shot	0.380 [.351, .407]	0.322 [.292, .348]	0.370	0.342	0.427	0.247	0.156
EuroLLM-9B, five-shot	0.395 [.365, .425]	0.312 [.285, .341]	0.386	0.348	0.456	0.248	0.103
PEFT-en	0.385 [.351, .419]	0.337 [.303, .372]	0.367	0.426	0.351	0.199	0.218
PEFT-hr	0.403 [.369, .434]	0.359 [.325, .393]	0.380	0.426	0.382	0.201	0.227
PEFT-bilingual	0.436 [.403, .467]	0.388 [.355, .421]	0.420	0.434	0.439	0.202	0.247
Qwen3.5-9B, zero-shot	0.422 [.389, .456]	0.375 [.341, .408]	0.408	0.471	0.383	0.186	0.260
Qwen3.5-9B, five-shot	**0.444** [.414, .473]	0.349 [.322, .376]	0.438	0.386	0.524	0.233	0.120

6.2 Seeds and Learning Curve

Over seeds 42, 67 and 128 (Table 2), the standard deviations of the PEFT conditions were at most 0.018 in micro-F1, the largest being PEFT-bilingual on Croatian. The averages keep the ordering above with smaller margins: PEFT-bilingual still scored highest in Croatian (0.416 against 0.403 for PEFT-hr), a lead of about one standard deviation, and PEFT-en and PEFT-bilingual remained level in English (0.504 and 0.505).

Table 2: PEFT on test, mean ± standard deviation over seeds 42, 67 and 128.

System	Lang	micro-F1	Jaccard	macro-F1
PEFT-en	en	0.504 ± 0.012	0.434 ± 0.006	0.489 ± 0.017
PEFT-en	hr	0.385 ± 0.003	0.339 ± 0.002	0.366 ± 0.001
PEFT-hr	hr	0.403 ± 0.000	0.354 ± 0.004	0.379 ± 0.002
PEFT-bilingual	en	0.505 ± 0.007	0.442 ± 0.009	0.490 ± 0.005
PEFT-bilingual	hr	0.416 ± 0.018	0.364 ± 0.023	0.400 ± 0.018

On the learning curve (Figure 1), PEFT-bilingual scored above PEFT-hr at all five sizes, by between 0.006 and 0.069 in micro-F1. With 250 Croatian lines, PEFT-hr (0.343) fell below zero-shot prompting, while PEFT-bilingual (0.388) was level with PEFT-en, and with 1,000 lines PEFT-bilingual came within 0.003 of its full-data score. The curve of PEFT-hr is not monotonic (0.380, 0.365 and 0.416 with 500, 1,000 and 2,000 lines). Each point is a single run.

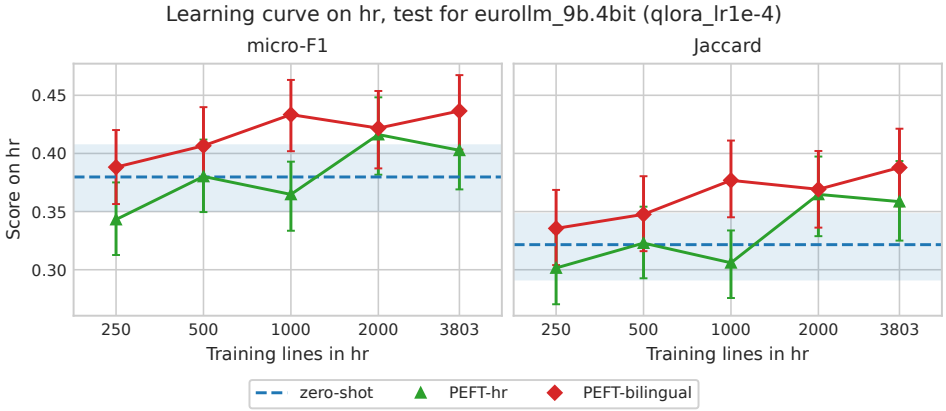


Figure 1: Test scores on Croatian against the number of Croatian training lines, with 95% bootstrap intervals; the dashed line and band are zero-shot prompting.

6.3 Per-Emotion Results

The systems differed more between emotions than from each other (Figure 2). Anger and joy were the easiest emotions in both languages, and disgust was weak for every system, with an F1 between 0.15 and 0.38. In English, fine-tuning improved every emotion over zero-shot prompting, sadness most of all (from 0.26 to 0.49 F1 for PEFT-en; Figure 12). In Croatian, the gains were smaller, with PEFT-bilingual improving six of the eight emotions by between 0.03 and 0.10 but not fear or disgust.

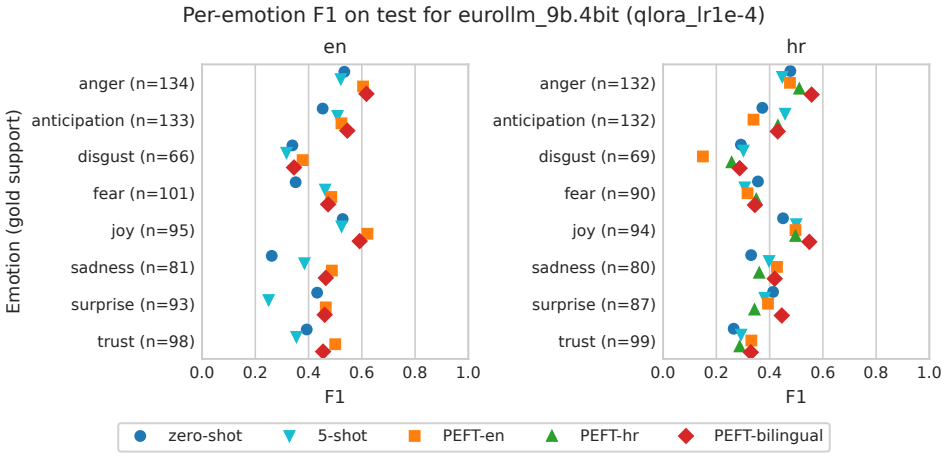


Figure 2: Per-emotion F1 on test, with the gold support of each emotion.

7 Analysis and Discussion

7.1 Cross-Lingual Transfer

Fine-tuning on English alone did not carry over to Croatian: PEFT-en scored about the same as zero-shot prompting on Croatian and had the largest gap between the two languages of any system. In English, fine-tuning raised both precision and recall over zero-shot prompting, but in Croatian it raised precision from 0.34 to 0.43 while lowering recall from 0.43 to 0.35, and PEFT-en assigned a single emotion to more than four in five Croatian lines, against about two in three in the reference labels (Figure 3). English training thus taught the model to give fewer labels, and this habit carried over to Croatian, but its better recognition of the emotions themselves largely did not. PEFT-hr showed the same pattern less strongly.

Croatian training data closed part of the gap, and combining it with English worked best. Over three seeds, PEFT-bilingual led PEFT-hr by only about one standard deviation, but the learning curve gives firmer evidence: PEFT-bilingual scored above PEFT-hr at every size of the Croatian training set, and with only 250 Croatian lines it stayed above zero-shot prompting while PEFT-hr fell below it. This advantage did not come from longer training, since the PEFT-bilingual checkpoint was kept after half an epoch, about a third of the optimisation steps of the PEFT-hr checkpoint. The English lines therefore appear to teach the task, which a modest amount of Croatian data then adapts to the language.

Pretrained language coverage did not predict which model scored higher. EuroLLM-9B was built for Croatian, whereas Qwen3.5-9B does not document whether it covers it, yet Qwen3.5-9B scored higher in Croatian with both prompts. EuroLLM-9B did lose less from English to Croatian when prompted (0.048 in micro-F1 zero-shot and 0.030 five-shot, against 0.062 and 0.069 for Qwen3.5-9B), which fits its explicit coverage of Croatian, but the two models also differ in other ways, including their numeric precision, so this comparison cannot isolate the effect of coverage.

7.2 Label Counts and Output Format

How many labels a system gave per line mattered more for its scores than producing the required format (Figure 3). On the test set, the reference labels give a single emotion to about two thirds of the

lines, with 1.42 labels per Croatian line on average. Zero-shot EuroLLM-9B gave 1.77 per Croatian line, and five-shot prompting gave two labels to 70–74% of the lines in both languages, in line with its demonstrations. This over-labelling explains why five-shot prompting raised recall but lowered precision and the exact-match ratio (Table 1). Fine-tuning brought the number of labels close to that of the reference, with 1.44 labels per Croatian line for PEFT-bilingual, which accounts for much of its gain; only PEFT-en and PEFT-hr gave too few on Croatian.

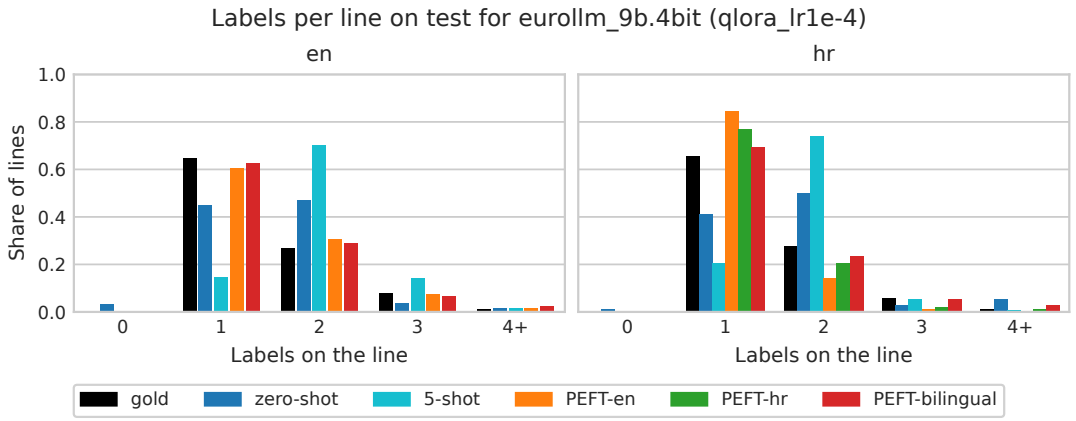


Figure 3: Number of labels per line in the gold labels and in each system's predictions on test.

Apart from the single malformed answer, every answer was a valid JSON list. Empty answers occurred only in zero-shot prompting, at most 3.3% (Figure 11). Every empty Croatian answer of EuroLLM-9B named an emotion outside the label set, most often `"disappointment"`, which the parser does not map to a label.

7.3 Per-Emotion Behaviour

Prompted EuroLLM-9B over-predicted disgust, the rarest emotion. It found more than half of the disgust lines, but only about one in five of its disgust predictions was correct, and fine-tuning, which had the fewest disgust examples to learn from, raised this precision to at most 0.30.

The five demonstrations changed micro-F1 little because their effects on single emotions pulled in opposite directions. In English, five-shot prompting raised the F1 for fear and sadness by about 0.11 and 0.12 but lowered it for surprise by 0.18, as recall for surprise fell from 0.46 to 0.18, even though two of the five demonstrations carry surprise.

7.4 Prompt Sensitivity

We measured sensitivity to the prompt on the development set only, using the three instruction templates and, for five-shot prompting, a random set of demonstrations in place of the coverage set (Table 3; all runs in Table 9 and Figure 10). Rewording the instruction changed micro-F1 by at most 0.038. In zero-shot prompting, Croatian was about twice as sensitive as English for both models, and the terse template scored lowest on Croatian, leaving 8.4% of the Croatian answers of Qwen3.5-9B empty, against 1.9% with the default template. Demonstrations reduced the sensitivity to wording, most of all for Qwen3.5-9B.

For EuroLLM-9B, the coverage demonstrations scored 0.025 (English) and 0.037 (Croatian) above random ones, more than rewording changed its five-shot scores, while for Qwen3.5-9B the two sets

Table 3: Prompt sensitivity on dev: mean, standard deviation and range of micro-F1 across the three instruction templates. Five-shot rows use the coverage demonstrations.

Model	Shots	Lang	Mean	SD	Range
EuroLLM-9B	0	en	0.428	0.006	0.015
EuroLLM-9B	0	hr	0.377	0.013	0.031
EuroLLM-9B	5	en	0.438	0.007	0.015
EuroLLM-9B	5	hr	0.412	0.008	0.019
Qwen3.5-9B	0	en	0.480	0.007	0.017
Qwen3.5-9B	0	hr	0.410	0.016	0.038
Qwen3.5-9B	5	en	0.521	0.004	0.007
Qwen3.5-9B	5	hr	0.467	0.002	0.005

were within 0.006 of each other. With random demonstrations, both models gave fewer labels, with higher precision and lower recall. The coverage set carries about two emotions per line, against about 1.45 in the data, and few-shot predictions are known to be biased towards what the demonstrations show (Zhao et al., 2021), which suggests that our selection procedure explains part of the over-labelling of five-shot prompting. The default template, used for all test runs, was never behind the best variant by more than 0.014 in micro-F1, well within the development bootstrap intervals, so our conclusions do not depend on the wording.

7.5 PEFT versus Prompting: Cost

Fine-tuning bought its accuracy with a one-off training cost: each PEFT run took between 0.70 and 0.86 hours on a single A100 and used at most 13.8 GiB of GPU memory (Figure 6), while a full prompting pass over the development set took about four to six minutes. Since a fine-tuned model receives the same input as zero-shot prompting, it costs no more in use. In return, fine-tuning raised micro-F1 over zero-shot prompting by about 0.09 in English and by 0.056 in Croatian, or 0.036 for the mean over three seeds. The configuration search cost more (18 runs, about 13.6 GPU hours), and its main cost result is that QLoRA used about a third less GPU memory than LoRA (13.4 to 13.8 against 21.1 to 21.4 GiB) with no loss on the development set.

A stronger base model changes the balance. Five-shot Qwen3.5-9B matched or exceeded the fine-tuned EuroLLM-9B averaged over three seeds (0.513 against 0.505 in English, 0.444 against 0.416 in Croatian) without any training, although it ran unquantised rather than in 4-bit. Fine-tuning also needs labelled target-language data, although the learning curve suggests that about 1,000 Croatian lines alongside English are enough.

7.6 Error Analysis

To see where fine-tuning changed individual predictions, we compared zero-shot prompting and PEFT-bilingual on every Croatian test line by the per-line Jaccard index. PEFT-bilingual improved 150 lines (27%) and worsened 94 (17%); the other 307 scored equally, including 172 on which neither system found a correct label. Table 4 shows five illustrative lines; their aligned English lines help us read them but do not validate the labels.

On 116 of the 150 improved lines, PEFT-bilingual predicted fewer emotions that the reference does not contain; most improvements came from dropping wrong labels. It most often removed disgust (49 lines) and anger (24), as in row 1, where zero-shot prompting gave both to an agreeing “Yeah, absolutely”. The worsened lines were of two kinds. Some predictions are defensible readings that the reference does not share: in row 2, missing someone can express sadness as well as joy, and without the surrounding dialogue a single line often supports more than one reading. Others are plain model errors, such as row 3, where PEFT-bilingual listed all eight emotions.

Table 4: Five Croatian test lines with their aligned English line, the number of alignment hops, the Croatian reference labels and the predictions of zero-shot prompting and PEFT-bilingual.

#	Croatian	Aligned English	Hops	Reference	Zero-shot	PEFT-bilingual
1	Da, potpuno.	Yeah, absolutely.	1	joy	anger, disgust	joy
2	Nedostajao si nam!	We missed ya!	1	joy	joy	sadness
3	Zna to?	You know what?	2	surprise	surprise	all eight
4	to je to?	Sign this.	1	anger, anticipation, joy, surprise	surprise	surprise
5	ena je to jako teko prihvatila, ali sam joj objasnio… da nita ne osete.	- What is it?	2	anticipation, surprise	anger, sadness	sadness

Some errors lie in the reference labels rather than in the models. On 35 test lines (6.4%), the Croatian reference differs from the human labels of its own aligned English line, and on 11 of them the two share no emotion. Some pairs are not translations of each other at all. In row 4, the Croatian line, which a machine translator renders as “What is that?”, is aligned with “Sign this.”, and in row 5 a long Croatian sentence about a woman taking bad news hard inherited the labels of “What is it?”. The sadness predicted by PEFT-bilingual fits the Croatian sentence but counts as wrong. Every system, including zero-shot prompting, also scored a mean Jaccard between 0.02 and 0.05 lower on lines whose labels came through Danish than on directly aligned lines, with overlapping intervals (Figure 13). As zero-shot prompting saw no training labels, this gap lies in the lines or labels, not in training, although label mismatches were not more frequent on two-hop lines (16 of 329) than on direct ones (19 of 222).

7.7 Limitations

The largest limitation lies in the Croatian reference labels. Our Croatian scores measure agreement with the projected reference, not with how a Croatian reader would label the lines, and without a Croatian reader we could neither measure nor correct the affected lines; even 175 English lines (3.2%) carry projected labels. Our design therefore cannot tell whether the lower Croatian scores reflect the models’ weaker command of Croatian, noise in the projected labels, or both, and it provides no evidence about cultural differences in how emotions are expressed, so we make no claims about them.

Our conclusions also have a limited scope: one language, one primary model of one size, and a second model that was only prompted, at a different precision, all classifying lines without their surrounding dialogue. With 551 test lines per language, the bootstrap intervals on micro-F1 span about ± 0.03, so small differences, such as between the two fine-tuned conditions in English, are within noise. The variation between training runs was measured with only three seeds, the learning curve and prompt variants are single runs, and with only 66 to 134 test lines per emotion we computed no per-emotion intervals and draw per-emotion conclusions only from patterns that hold across systems or languages. Finally, PEFT-hr was still improving slightly when it reached the three-epoch limit, and longer training might narrow its gap to PEFT-bilingual.

8 Conclusion and Future Work

With the same model, EuroLLM-9B, fine-tuning clearly outperformed prompting in English, raising micro-F1 from 0.428 with zero-shot prompting to 0.517, and in Croatian the best result came from bilingual fine-tuning (0.436 against 0.380). Prompting gave too many labels, especially with five demonstrations, and much of the gain from fine-tuning came from learning how many emotions to report. Fine-tuning on English alone did not transfer to Croatian, but adding Croatian data to English helped most. Bilingual fine-tuning scored above Croatian-only fine-tuning at every amount of Croatian data, and came close to its full-data score with 1,000 Croatian lines.
Prompt wording mattered little next to the differences between systems, and five-shot Qwen3.5-9B matched the fine-tuned EuroLLM-9B without training, although it ran unquantised. All Croatian results, however, measure agreement with projected labels, some inherited from misaligned lines.
With more time, we would first have a Croatian reader verify the test labels, separating model errors from label errors. Beyond this, we see the following directions:

- **Translate-test evaluation:** Classifying machine-translated Croatian test lines with PEFT-en would show whether the Croatian gap lies in the model or in the labels.
- **Filtering misaligned pairs:** Rescoring every system on the pairs that a multilingual sentence-embedding model judges well aligned would estimate the projection noise without a Croatian reader.
- **Controlling the number of labels:** Thresholding the model's probability for each label instead of generating a free-form list, and choosing demonstrations whose label counts match the data, would target the over-labelling behind much of the difference between systems.
- **Dialogue context and a second fine-tuned model:** Adding preceding subtitle lines would address ambiguous lines, and fine-tuning both models at the same precision would show whether fine-tuning still pays off on a stronger base model.

Comparing our results with those for other resource bands would, in addition, show whether the gap between fine-tuning and prompting depends on the amount of target-language data.

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A Code Repository

The code is available at https://github.com/squidistaken/large-language-models, on the `main` branch. The version submitted with this report is published as a release on the repository's releases page. Each fine-tuned adapter has a model card in the `model_cards/` folder, describing its base model, training data, prompt template, settings and the commands to reproduce it. The adapter weights themselves are not published.

B Language Resource Bands

Table 5 lists the five resource bands that group XED's projected languages by their approximate number of annotated lines; the example languages are illustrative rather than exhaustive. Our assigned language, Croatian, is in band 3.

Table 5: XED resource bands and example languages.

Band	Resource level	Annotated lines	Example languages
1	extreme low	under ∼1,000	Macedonian (300), Vietnamese (956), Slovak (975), Icelandic (977)
2	very low	∼1,000–4,400	Danish, Russian, Bosnian, Slovenian, Arabic, Norwegian
3	mid-low	∼4,400–6,000	Hebrew, Swedish, Dutch, German, Hungarian, Croatian
4	mid-high	∼6,000–7,300	Czech, Italian, Bulgarian, Polish, Portuguese, French
5	high	∼8,000 and above	Greek, Finnish, Serbian, Turkish, Romanian, Spanish, Brazilian Portuguese

C Split Statistics

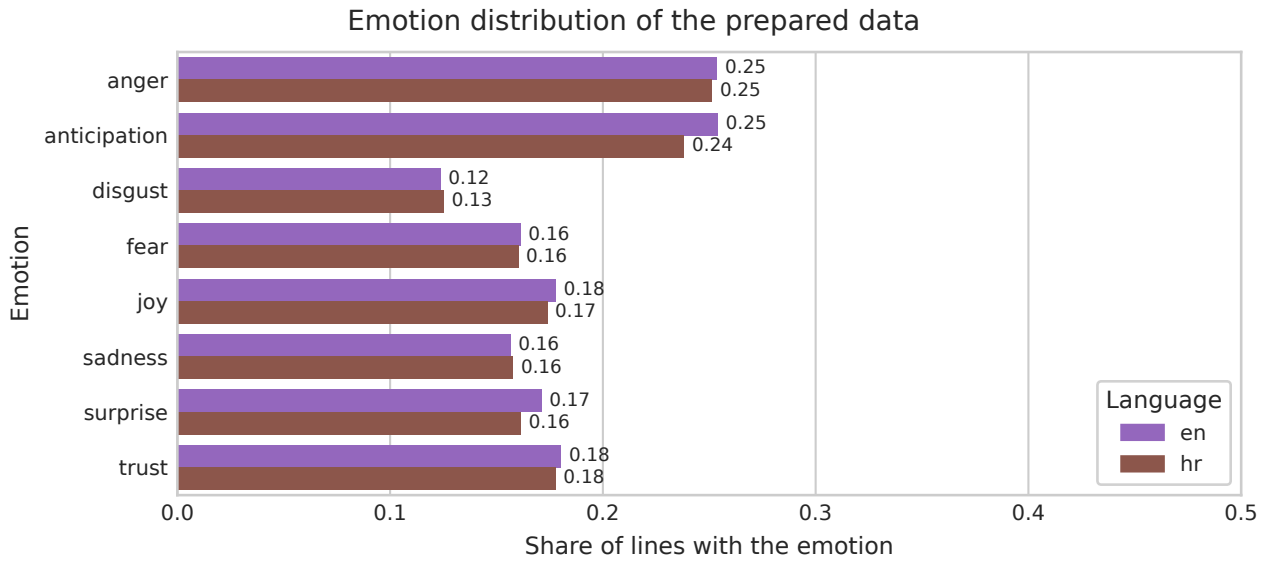


Figure 4: Share of lines carrying each emotion in the prepared English and Croatian data. The two languages are almost identical, as nearly all labels share one origin.

Table 6: Lines carrying each emotion per Croatian partition, with their share of the partition in parentheses. The split was stratified on these labels; the English partitions differ by at most 2.7 percentage points (fear).

Emotion	Train (3,803)	Dev (1,068)	Test (551)
anger	962 (25.3%)	269 (25.2%)	132 (24.0%)
anticipation	905 (23.8%)	255 (23.9%)	132 (24.0%)
disgust	478 (12.6%)	132 (12.4%)	69 (12.5%)
fear	610 (16.0%)	169 (15.8%)	90 (16.3%)
joy	663 (17.4%)	186 (17.4%)	94 (17.1%)
sadness	596 (15.7%)	180 (16.9%)	80 (14.5%)
surprise	620 (16.3%)	169 (15.8%)	87 (15.8%)
trust	665 (17.5%)	200 (18.7%)	99 (18.0%)

D Prompts

Every prompt has a system message and a user message. The system message is one of the instruction templates below, in which `{definitions}` is replaced by the label definitions; the user message holds the subtitle. In the five-shot prompts, the demonstrations (Table 7) are inserted as user and assistant turns between the two.

Table 7: Five fixed demonstrations of the five-shot prompt, shown in English for English evaluation and in Croatian for Croatian evaluation, each with its own labels.

#	English	Croatian	Labels (en)	Labels (hr)
1	- Think I should go out with Paulie?	- Hoe da izaem sa Paulijem?	anticipation, surprise, trust	anticipation, surprise, trust
2	Relax.	Smiri se.	anger, fear, joy	anger, fear
3	You got old.	Ostarili ste.	disgust, sadness	disgust, sadness
4	He said no pilots.	Nee pilota.	sadness, surprise	sadness, surprise
5	Shoot the fuck down!	Skini ih jebeno dolje!	fear	fear

Label definitions, replacing `{definitions}`

```
- anger: displeasure or hostility at a perceived wrong
- anticipation: expecting or looking forward to something
- disgust: revulsion or strong disapproval
- fear: feeling threatened or afraid
- joy: happiness or pleasure
- sadness: sorrow, loss or disappointment
- surprise: reaction to something unexpected
- trust: confidence in, or acceptance of, someone or something
```

User message

```
Subtitle: {text}
```

Template `default` (used on test)

```
You classify the emotions expressed in a movie subtitle line.

Permitted labels:
{definitions}

Every line expresses at least one of these emotions and may express
several. Answer with a JSON list of the matching label names, using
only the names above, and nothing else, in the format ["label", ...].
```

Template `terse` (dev only)

```
Label the emotions in this subtitle line. Choose one or more of:
{definitions}

Reply with only a JSON list of the chosen label names, like ["label",
...].
```

Template `annotator` (dev only)

```
You are an expert annotator of emotions in film dialogue, using
Plutchik's eight basic emotions:
{definitions}

Read the subtitle line and decide which of these emotions it
expresses; a line can express one emotion or several at once. Respond
with a JSON list of the label names exactly as written above, and no
other text.
```

E Fine-Tuning Configuration

E.1 Configuration Search

Table 8: Candidate configurations on the development set. The selection score averages dev micro-F1 per language over the conditions evaluated in it, then over the two languages.

Configuration	Base model, learning rate	Dev micro-F1 (en)	Dev micro-F1 (hr)	Selection score
qlora_lr1e-4	**4bit**, **1e-4**	**0.521**	**0.437**	**0.479**
qlora	4bit, 2e-4	0.517	0.438	0.477
lora_lr1e-4	bf16, 1e-4	0.512	0.439	0.475
lora_lr4e-4	bf16, 4e-4	0.510	0.436	0.473
qlora_lr4e-4	4bit, 4e-4	0.509	0.437	0.473
lora	bf16, 2e-4	0.510	0.431	0.470

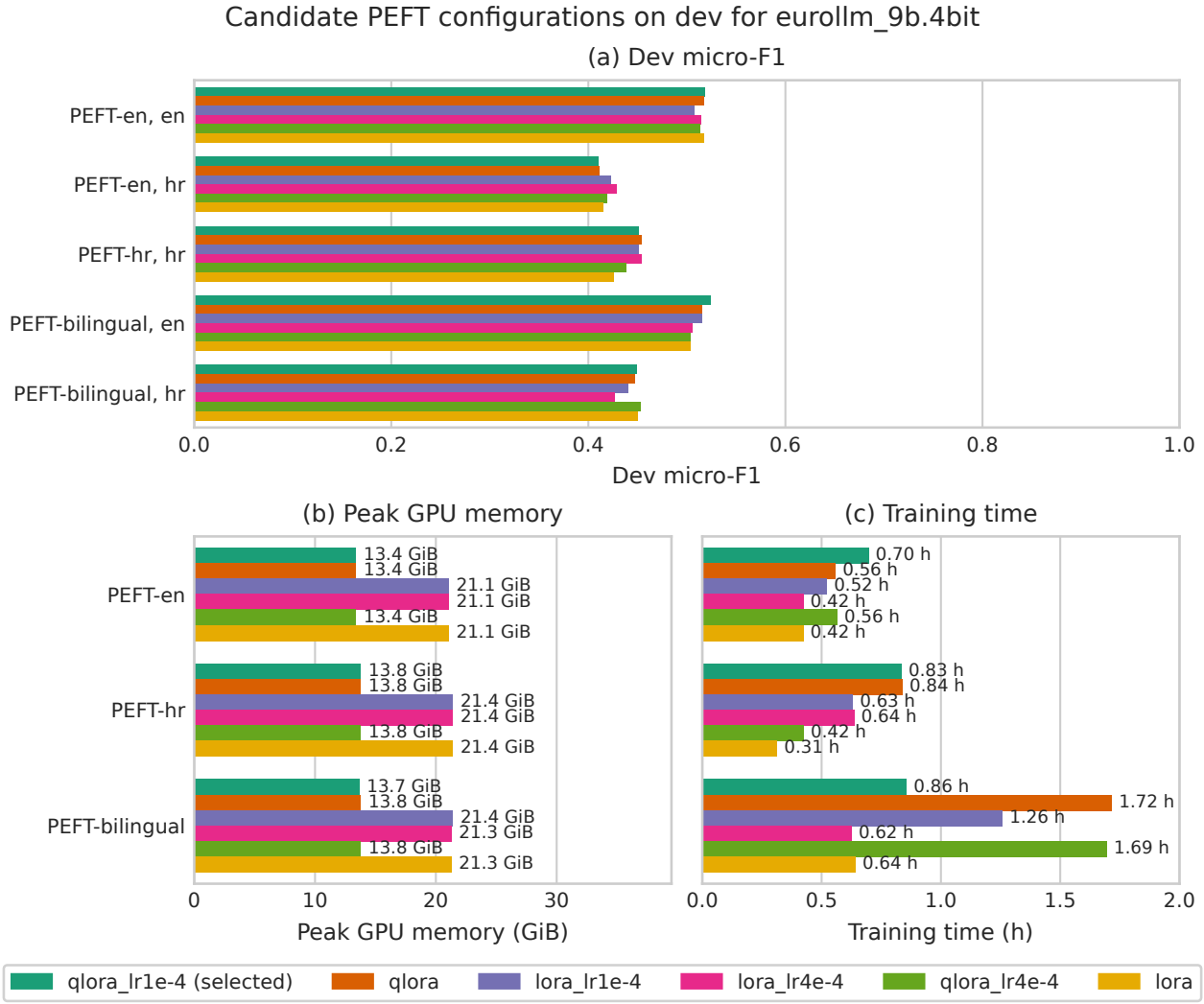


Figure 5: Dev micro-F1, peak GPU memory and training time of every candidate configuration.

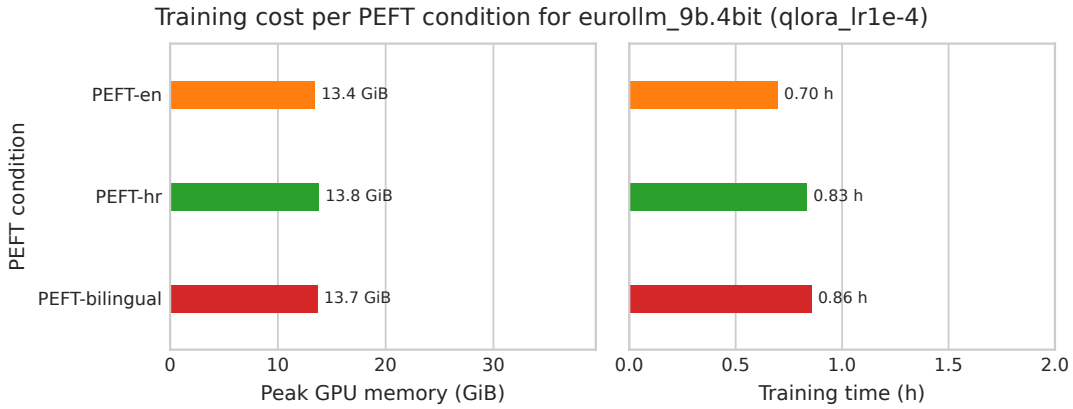


Figure 6: Peak GPU memory and training time of each PEFT condition of the selected configuration.

E.2 Training Histories

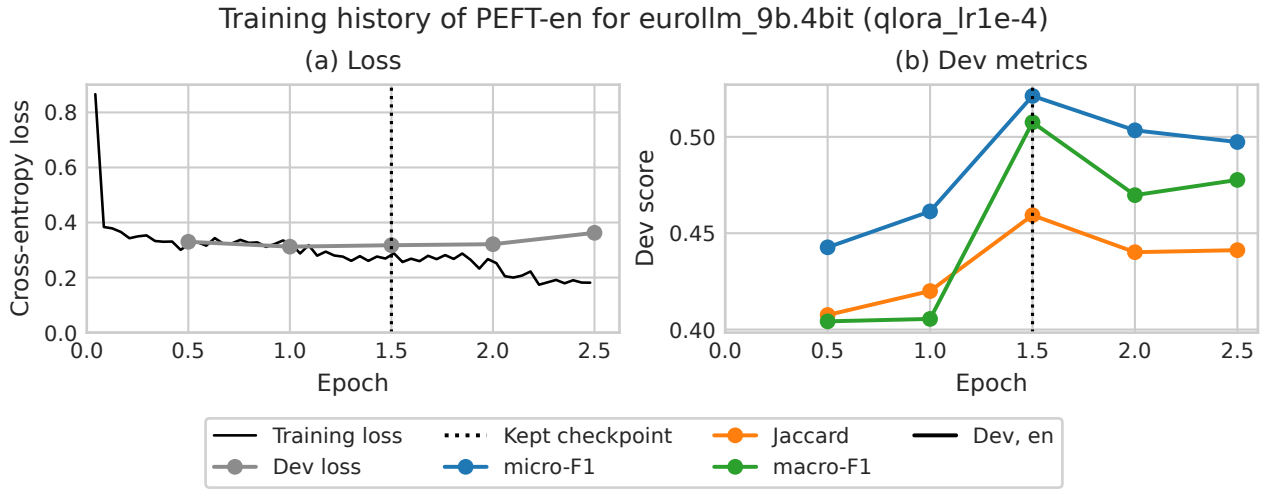


Figure 7: Training history of PEFT-en in the selected configuration.

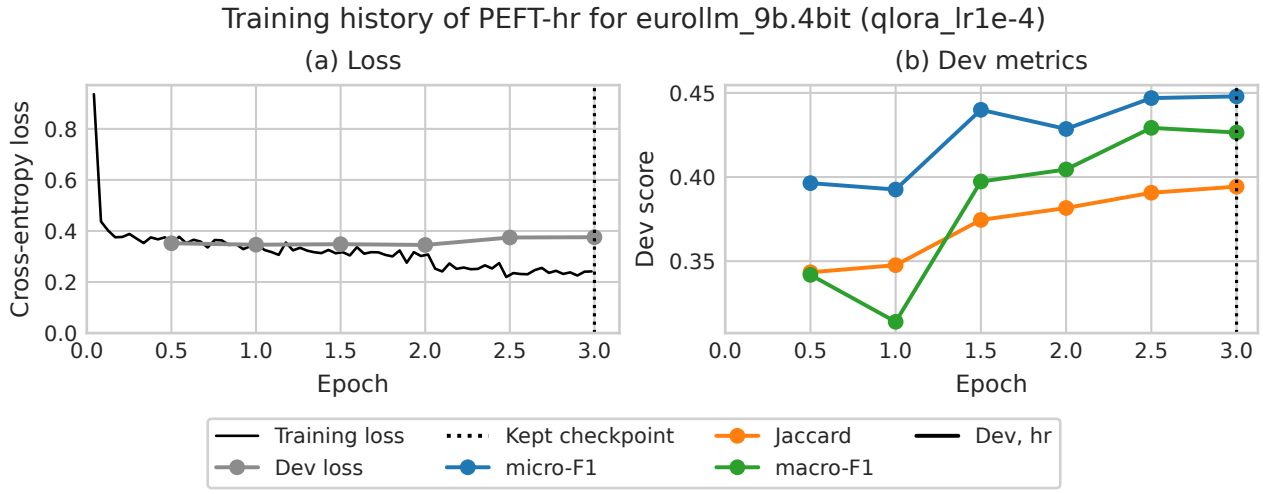


Figure 8: Training history of PEFT-hr in the selected configuration.

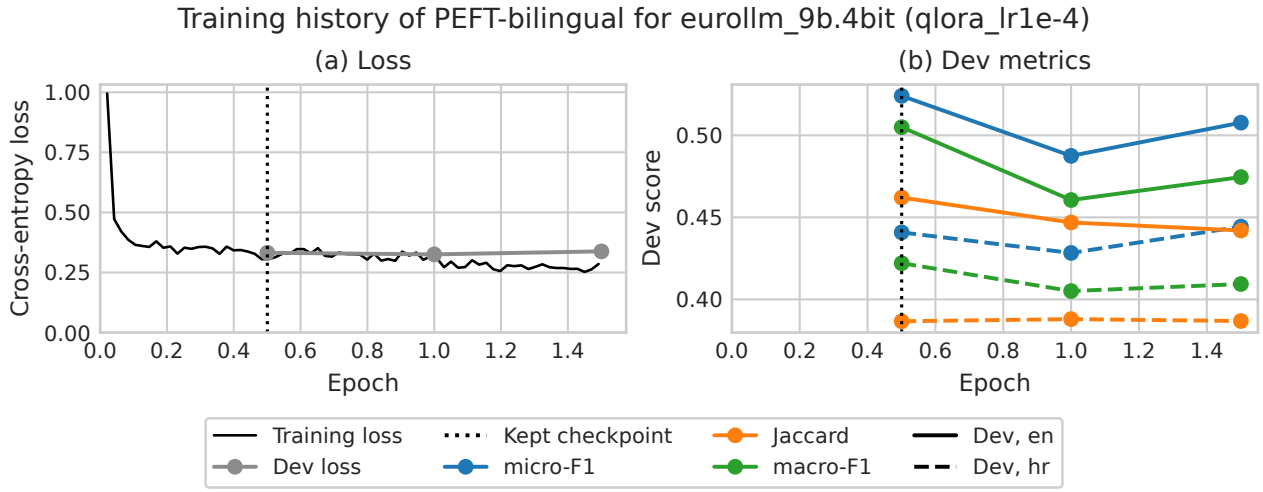


Figure 9: Training history of PEFT-bilingual in the selected configuration.

F Prompting on the Development Set

Table 9: Every prompting run on dev (micro-F1 and Jaccard).

Model	Shots	Variant	Lang	micro-F1	Jaccard
EuroLLM-9B	0	default	en	0.430	0.362
EuroLLM-9B	0	default	hr	0.381	0.312
EuroLLM-9B	0	annotator	en	0.435	0.335
EuroLLM-9B	0	annotator	hr	0.391	0.298
EuroLLM-9B	0	terse	en	0.420	0.335
EuroLLM-9B	0	terse	hr	0.360	0.282
EuroLLM-9B	5	default	en	0.441	0.346
EuroLLM-9B	5	default	hr	0.419	0.331
EuroLLM-9B	5	annotator	en	0.429	0.340
EuroLLM-9B	5	annotator	hr	0.400	0.313
EuroLLM-9B	5	random	en	0.416	0.355
EuroLLM-9B	5	random	hr	0.382	0.321
EuroLLM-9B	5	terse	en	0.445	0.344
EuroLLM-9B	5	terse	hr	0.416	0.320
Qwen3.5-9B	0	default	en	0.478	0.431
Qwen3.5-9B	0	default	hr	0.413	0.363
Qwen3.5-9B	0	annotator	en	0.490	0.431
Qwen3.5-9B	0	annotator	hr	0.427	0.362
Qwen3.5-9B	0	terse	en	0.473	0.421
Qwen3.5-9B	0	terse	hr	0.389	0.324
Qwen3.5-9B	5	default	en	0.519	0.426
Qwen3.5-9B	5	default	hr	0.466	0.369
Qwen3.5-9B	5	annotator	en	0.526	0.414
Qwen3.5-9B	5	annotator	hr	0.467	0.366
Qwen3.5-9B	5	random	en	0.517	0.446
Qwen3.5-9B	5	random	hr	0.460	0.394
Qwen3.5-9B	5	terse	en	0.518	0.418
Qwen3.5-9B	5	terse	hr	0.470	0.368

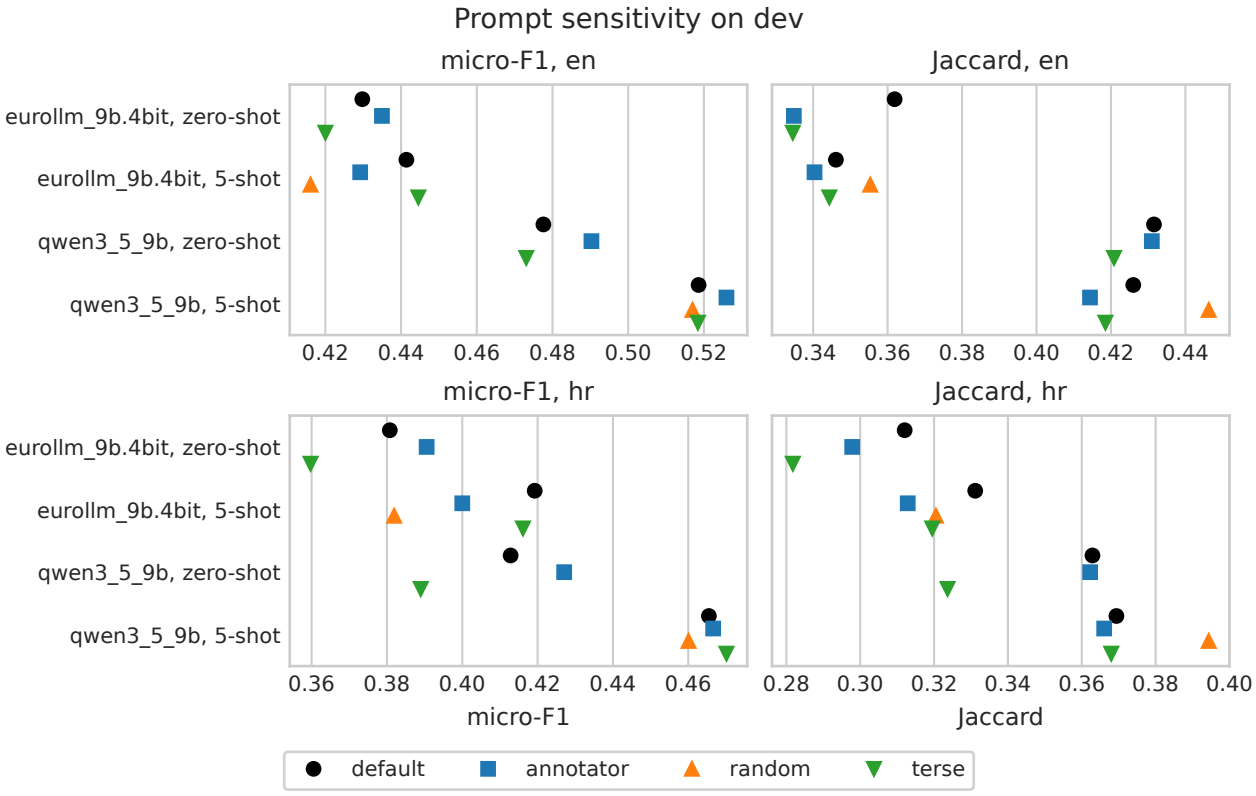


Figure 10: Micro-F1 and Jaccard on the development set for every prompt variant: the default prompt (black), the two reworded templates, and random instead of emotion-covering demonstrations.

G Further Test Results

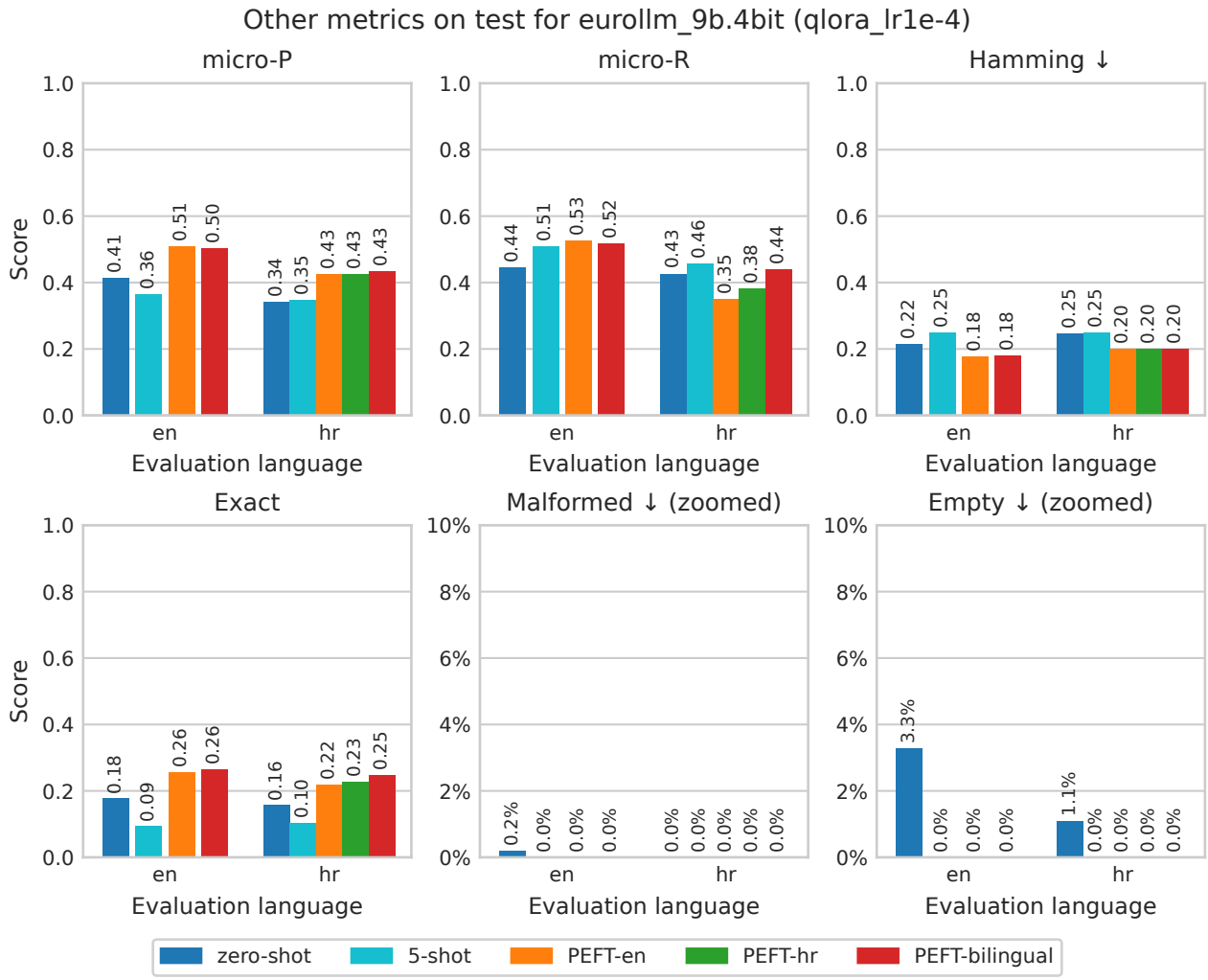


Figure 11: Secondary and diagnostic metrics on test, complementing Table 1; ↓ marks metrics for which lower is better, and the malformed and empty rates are on a zoomed scale.

Table 10: Per-emotion precision / recall / F1 on the English test set, with the gold support of each emotion. E-0 and E-5 are zero- and five-shot EuroLLM-9B, Q-0 and Q-5 zero- and five-shot Qwen3.5-9B, and PEFT-bi is PEFT-bilingual.

Emotion	*n*	E-0	E-5	PEFT-en	PEFT-bi	Q-0	Q-5
anger	134	.52/.55/.53	.54/.51/.52	.54/.68/.60	.54/.72/.62	.59/.59/.59	.63/.65/.64
anticipation	133	.37/.58/.45	.48/.54/.51	.54/.50/.52	.49/.61/.55	.59/.29/.39	.52/.41/.46
disgust	66	.24/.58/.34	.20/.74/.32	.30/.52/.38	.28/.44/.35	.33/.15/.21	.28/.41/.33
fear	101	.45/.29/.35	.51/.43/.46	.60/.41/.49	.59/.40/.47	.54/.49/.51	.45/.63/.52
joy	95	.57/.49/.53	.40/.75/.52	.62/.62/.62	.57/.61/.59	.59/.67/.63	.48/.75/.58
sadness	81	.54/.17/.26	.29/.56/.38	.49/.48/.49	.54/.41/.46	.58/.40/.47	.48/.62/.54
surprise	93	.41/.46/.43	.40/.18/.25	.58/.39/.46	.53/.41/.46	.47/.45/.46	.41/.59/.49
trust	98	.45/.35/.39	.30/.43/.35	.46/.55/.50	.51/.41/.45	.49/.37/.42	.35/.76/.48

Table 11: Per-emotion precision / recall / F1 on the Croatian test set, with the gold support of each emotion. E-0 and E-5 are zero- and five-shot EuroLLM-9B, Q-0 and Q-5 zero- and five-shot Qwen3.5-9B, and PEFT-bi is PEFT-bilingual.

Emotion	*n*	E-0	E-5	PEFT-en	PEFT-hr	PEFT-bi	Q-0	Q-5
anger	132	.41/.58/.48	.65/.34/.45	.46/.50/.48	.46/.58/.51	.51/.62/.56	.54/.48/.51	.52/.54/.53
anticipation	132	.35/.39/.37	.34/.68/.46	.45/.27/.34	.40/.46/.43	.40/.47/.43	.49/.20/.29	.43/.43/.43
disgust	69	.20/.55/.29	.21/.51/.30	.21/.12/.15	.27/.25/.26	.26/.32/.29	.44/.17/.25	.24/.35/.28
fear	90	.37/.34/.36	.38/.26/.31	.42/.26/.32	.40/.31/.35	.45/.28/.34	.41/.39/.40	.35/.56/.43
joy	94	.39/.53/.45	.41/.64/.50	.49/.50/.50	.51/.49/.50	.54/.56/.55	.49/.59/.53	.44/.62/.51
sadness	80	.45/.26/.33	.33/.50/.40	.40/.46/.43	.41/.33/.36	.40/.44/.42	.48/.46/.47	.44/.53/.48
surprise	87	.35/.51/.41	.45/.33/.38	.58/.30/.39	.49/.26/.34	.47/.43/.45	.45/.44/.44	.35/.46/.40
trust	99	.33/.22/.27	.25/.35/.29	.34/.32/.33	.45/.21/.29	.39/.28/.33	.42/.33/.37	.33/.69/.45

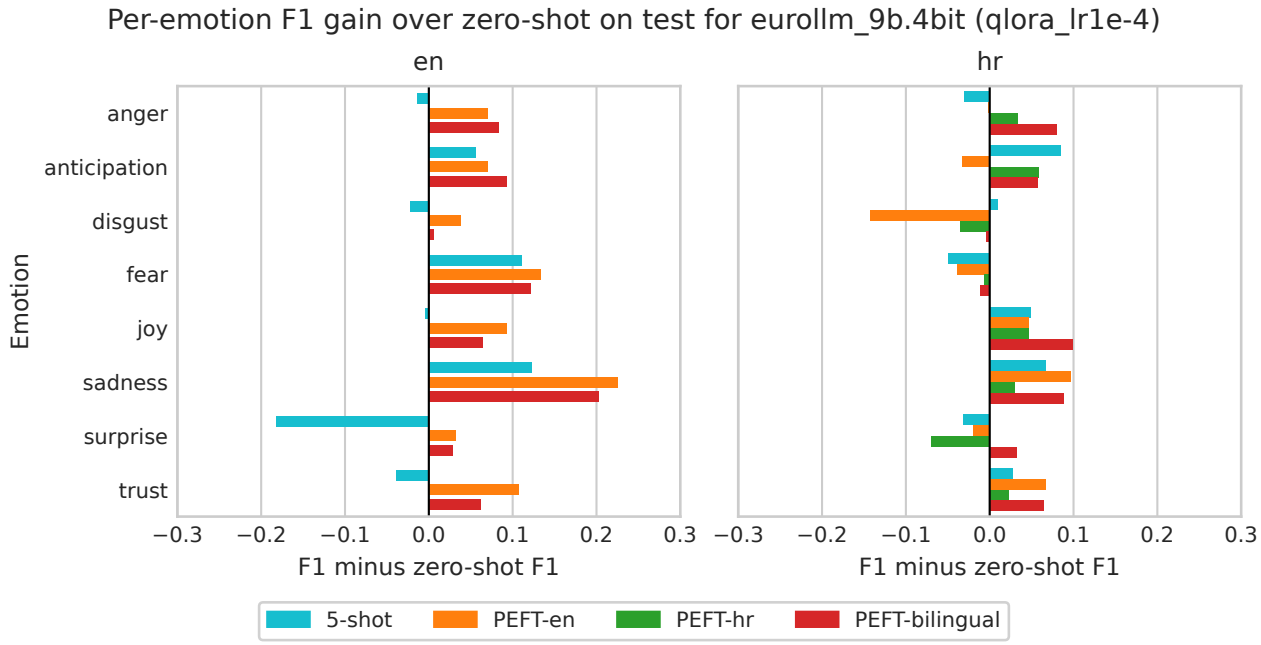


Figure 12: Per-emotion F1 gain of each system over zero-shot on test.

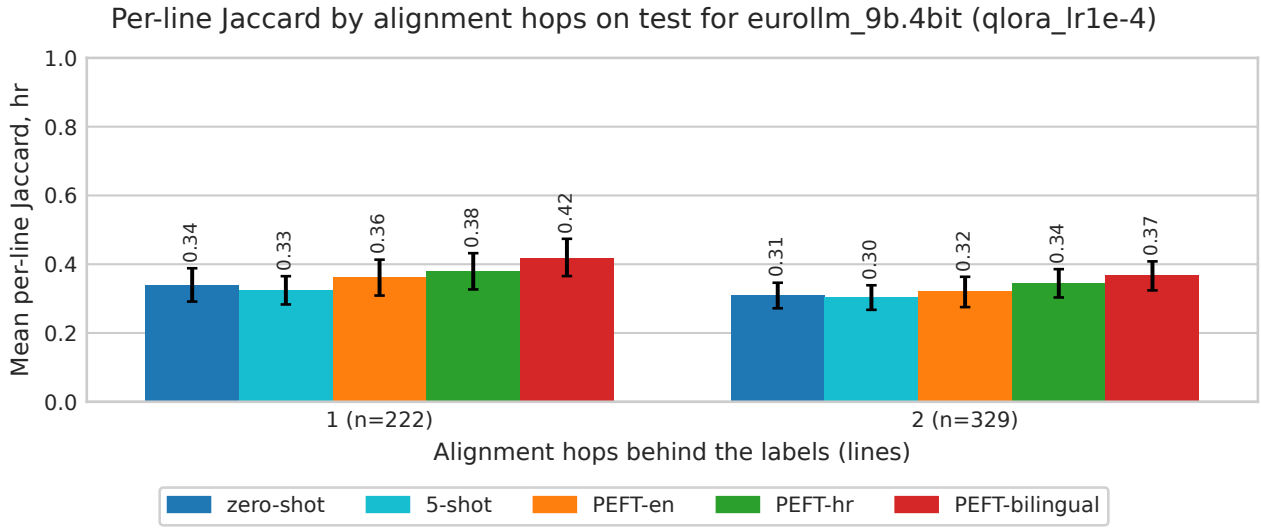


Figure 13: Mean per-line Jaccard on Croatian test lines by the number of alignment hops behind their projected labels, with 95% bootstrap intervals.